

Interpretable Recurrent Neural Network Baselines for Daily XAUUSD Direction Forecasting

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Abstract—This paper investigates recurrent neural network baselines for daily directional forecasting of XAUUSD using a prepared multivariate feature set of technical and return-based indicators. The study uses 2,715 daily observations spanning March 16, 2015 to December 30, 2025, with 25 numeric predictors and a binary target. A strict chronological protocol is adopted to reduce look-ahead bias: 70% of the data are used for training, 15% for validation, and 15% for out-of-sample testing. Each sample is represented as a rolling 20-day feature window, yielding 2,696 sequences. Two sequence classifiers are evaluated, namely a long short-term memory (LSTM) network and a gated recurrent unit (GRU) network. The GRU obtains the highest validation ROC AUC of 0.5488 and a test ROC AUC of 0.5570, while the LSTM achieves a slightly higher test accuracy of 0.5760. SHapley Additive exPlanations (SHAP) reveal that lagged returns and momentum-related variables, especially `return_lag_2`, `macd`, `macd_sig`, and `macd_hist`, dominate the learned signal. The empirical results indicate that recurrent baselines extract a weak but non-random predictive structure from the engineered XAUUSD features; however, the modest AUC values also suggest that the task remains difficult and that stronger regularization, alternative architectures, and trading-aware evaluation are required before deployment.

Index Terms—XAUUSD, LSTM, GRU, sequence classification, SHAP explainable AI, financial time series.

I. INTRODUCTION

Directional forecasting in financial markets remains difficult because market prices exhibit high noise, regime shifts, and rapidly changing signal-to-noise ratios. Nevertheless, sequence models remain attractive for this setting because they can exploit temporal structure in engineered features without assuming independent observations. In particular, gated recurrent architectures such as LSTM and GRU were introduced to address the long-term dependency and gradient propagation limitations of vanilla recurrent neural networks [1].

For market prediction tasks, recurrent models have shown promise when the forecasting problem is framed as out-of-sample directional classification on chronologically ordered data [4]. At the same time, explainability is essential in financial applications because weak predictive edges can be unstable, and model transparency is needed to interpret whether a system is learning economically plausible structure rather than noise. SHAP provides a principled additive explanation framework for local and global feature attribution and is

therefore a practical tool for auditing deep models in time-series settings [5].

This paper documents a compact but reproducible experiment on daily XAUUSD directional classification. The contribution is threefold. First, it establishes interpretable LSTM and GRU baselines on a prepared multivariate XAUUSD dataset. Second, it evaluates the models using a leakage-aware chronological split and multiple classification metrics. Third, it analyzes the learned signal using SHAP to identify the most influential feature groups across the 20-day lookback window. The paper intentionally reports the observed weaknesses as well as strengths of the system, since honest characterization of model quality is necessary for scientific usefulness.

II. DATA AND EXPERIMENTAL DESIGN

A. Dataset

The study uses the repository file `xauusd_features.csv`, which contains 2,715 daily observations from March 16, 2015 through December 30, 2025. Each row contains a binary target and 25 engineered numeric predictors. The target is the direction label supplied with the dataset. The predictors are grouped in Table I.

The raw target distribution is mildly imbalanced rather than extreme: the positive class accounts for 53.11% of all observations. The test period is more upward-skewed at 58.58%, which makes ROC AUC and F1 more informative than accuracy alone.

TABLE I
FEATURE GROUPS USED IN THE STUDY

Group	Features
Return features	<code>return_1d</code> , <code>return_5d</code> , <code>return_10d</code> , <code>log_return</code>
Lagged returns	<code>return_lag_1</code> , <code>return_lag_2</code> , <code>return_lag_3</code> , <code>return_lag_5</code>
Trend features	<code>sma_5</code> , <code>sma_10</code> , <code>sma_20</code> , <code>sma_50</code> , <code>ema_12</code> , <code>ema_26</code>
Volatility features	<code>volatility_5</code> , <code>volatility_10</code> , <code>volatility_20</code>
Momentum and oscillator features	<code>rsi_14</code> , <code>macd</code> , <code>macd_sig</code> , <code>macd_hist</code> , <code>bb_pos</code>
Intraday and volume features	<code>high_low_range</code> , <code>close_open</code> , <code>volume_z</code>

B. Chronological Split and Sequence Construction

To avoid leakage, the data are split chronologically rather than randomly. The first 70% of rows are used for training, the next 15% for validation, and the final 15% for testing. Feature standardization is fitted on the training slice only and then applied to the full series. A rolling lookback window of length 20 transforms the data into supervised sequences of shape 20×25 . The resulting sample counts are summarized in Table II.

TABLE II
CHRONOLOGICAL DATA SPLITS

Split	Rows	Sequences	Date Range
Train	1,900	1,881	2015-04-13 to 2022-10-03
Validation	407	407	2022-10-04 to 2024-05-16
Test	408	408	2024-05-17 to 2025-12-30

C. Model Architectures

Two recurrent baselines are trained:

- LSTM: one LSTM layer with 64 hidden units, followed by a dense layer with 32 rectified linear units, dropout, and a sigmoid output layer.
- GRU: one GRU layer with 64 hidden units, followed by the same dense, dropout, and sigmoid output stack.

Both models use binary cross-entropy loss, the Adam optimizer, batch size 32, and a maximum of 40 epochs. Early stopping monitors validation AUC with patience 6 and restores the best weights. The binary decision threshold is 0.5.

D. Evaluation Metrics

The models are evaluated with accuracy, precision, recall, F1-score, and ROC AUC. Model selection is based on validation ROC AUC. In addition to scalar metrics, the experiment records ROC curves, training histories, confusion matrices, and SHAP-based explanation plots.

III. EXPLAINABILITY PROTOCOL

After model selection, the best recurrent model is reloaded from the saved bundle and analyzed using SHAP GradientExplainer [5]. The explainer uses up to 64 background samples and 128 test samples. Because the recurrent input tensor has a temporal dimension, each explanation is flattened into a time-feature representation of size $20 \times 25 = 500$ for global visualization. The study reports:

- a SHAP beeswarm summary plot,
- a mean absolute SHAP bar chart,
- dependence plots for the four highest-ranked features at the most recent time step, and
- a waterfall explanation for one correctly classified positive sample.

This design supports both global interpretation of feature importance and local interpretation of a single prediction.

IV. RESULTS

A. Predictive Performance

Table III reports the validation and test metrics produced by the implementation. The GRU achieves the best validation ROC AUC of 0.5488 and is therefore selected as the final model. On the test set, the GRU attains the highest ROC AUC of 0.5570, whereas the LSTM attains the highest accuracy of 0.5760. Both models produce very similar F1-scores near 0.634, indicating that the task is only weakly predictable from the available feature set.

TABLE III
VALIDATION AND TEST PERFORMANCE OF THE RECURRENT BASELINES

Model	Split	Accuracy	Precision	Recall	F1	ROC AUC
LSTM	Validation	0.4963	0.5211	0.6296	0.5702	0.5224
LSTM	Test	0.5760	0.6410	0.6276	0.6342	0.5539
GRU	Validation	0.5135	0.5333	0.6667	0.5926	0.5488
GRU	Test	0.5466	0.6015	0.6695	0.6337	0.5570

The difference between accuracy and AUC is worth noting. The test set contains a larger share of upward labels than the full sample, which partly benefits threshold-based accuracy. The AUC values therefore provide a more stable view of discriminative performance. From this perspective, both models perform only slightly above chance. These results are scientifically useful because they indicate that the current feature set and architecture capture some structure, but not enough to support strong claims of deployable forecasting power.

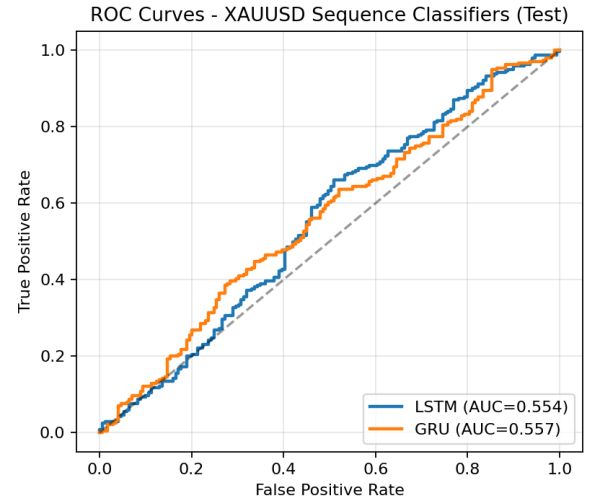


Fig. 1. Test-set ROC curves for the LSTM and GRU baselines. The GRU achieves the best out-of-sample ROC AUC, but both curves remain close to the diagonal, indicating limited separability.

Fig.1 shows the joint ROC comparison. The curves are consistently above the diagonal baseline, confirming nonrandom discrimination, but the margin is modest. This finding is consistent with the numerical AUC values in Table III.

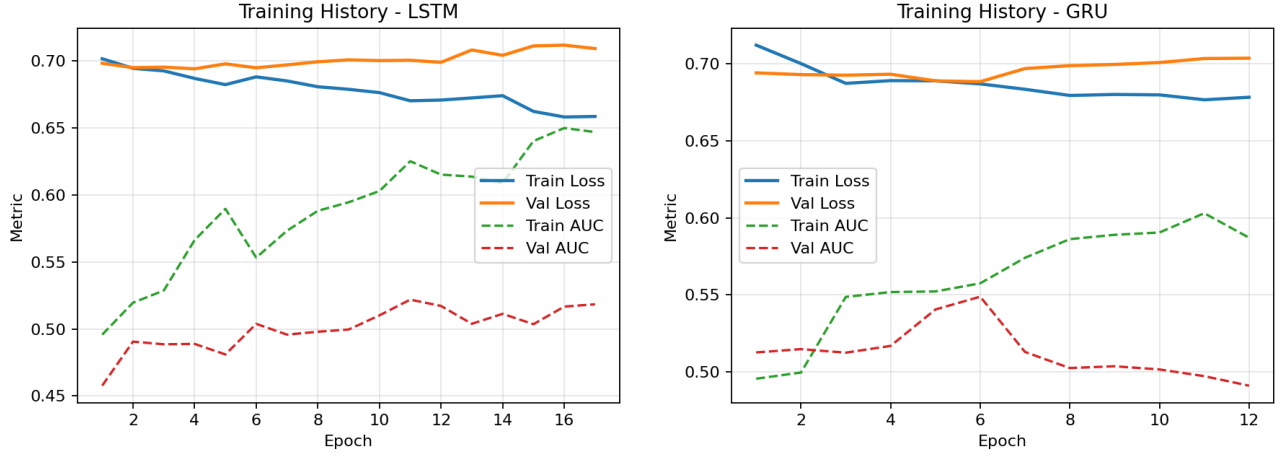


Fig. 2. Training histories of the two recurrent baselines. Both models converge rapidly and then plateau, which suggests that model capacity is sufficient for the current feature space but the underlying predictive signal remains weak.

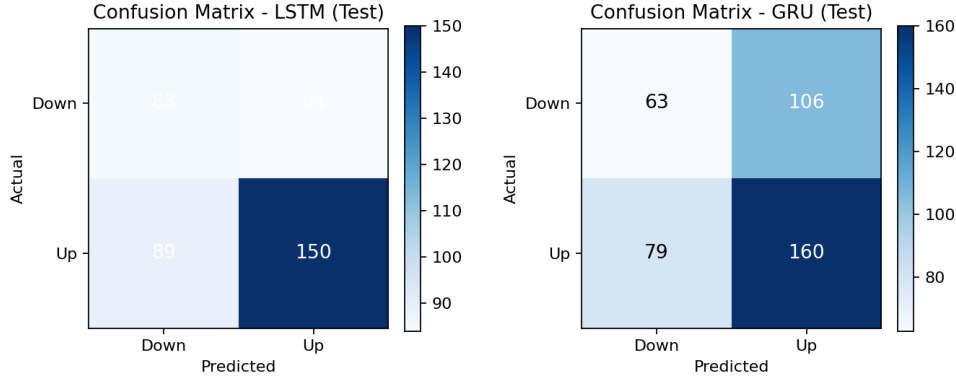


Fig. 3. Test-set confusion matrices for the recurrent baselines. The LSTM is more conservative than the GRU, while the GRU recovers a larger share of positive cases at the cost of more false positives.

The learning curves in 2 show stable optimization behavior with no catastrophic divergence. Validation performance saturates early, which supports the use of early stopping and also suggests that additional epochs alone are unlikely to close the generalization gap.

Fig. 3 further clarifies the behavior of the two classifiers. The GRU leans toward higher recall on the positive class, whereas the LSTM attains slightly stronger overall accuracy. This trade-off is aligned with the metric table and supports the paper’s choice to prioritize AUC for model selection rather than a single threshold-dependent statistic.

B. Global and Local Explainability

The SHAP analysis identifies a small set of features that dominate the learned signal. Table IV lists the top ten features after aggregating mean absolute SHAP values across all timesteps in the 20-day window.

The global SHAP summaries in Fig. 4 show that the explanatory mass is concentrated in a relatively small subset

of features. The strongest signals are dominated by short-lag return and momentum features rather than by long-window volatility or volume indicators. This concentration is a plausible outcome for a daily directional task in which the weak edge, if present, is likely to reside in recent reversal or trend continuation effects.

The dependence plots in Fig. 5 reinforce the conclusion that the model relies mainly on recent market state rather than distant history. In particular, `return_lag_2` appears as the single most influential variable, while `macd`, `macd_sig`, and `macd_hist` collectively indicate that momentum structure remains central to the model’s behavior.

The local explanation in Fig. 6 illustrates how a single prediction is assembled from additive contributions. The waterfall structure is consistent with the global results: only a small subset of recent inputs exerts substantial influence on the prediction. This is desirable from an interpretability standpoint because it suggests that the model is not spreading importance diffusely across all 500 time-feature positions.

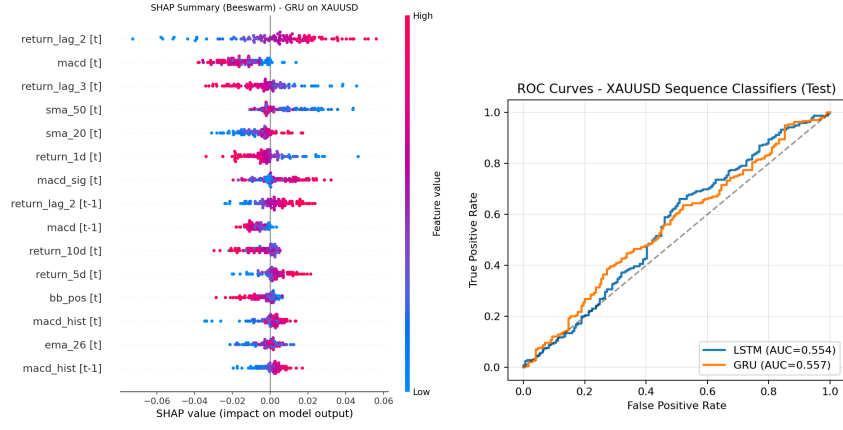


Fig. 4. Global SHAP summaries for the selected recurrent model. The explanations indicate that short-horizon lagged return information and MACD-derived momentum variables contribute most to the model's decisions.

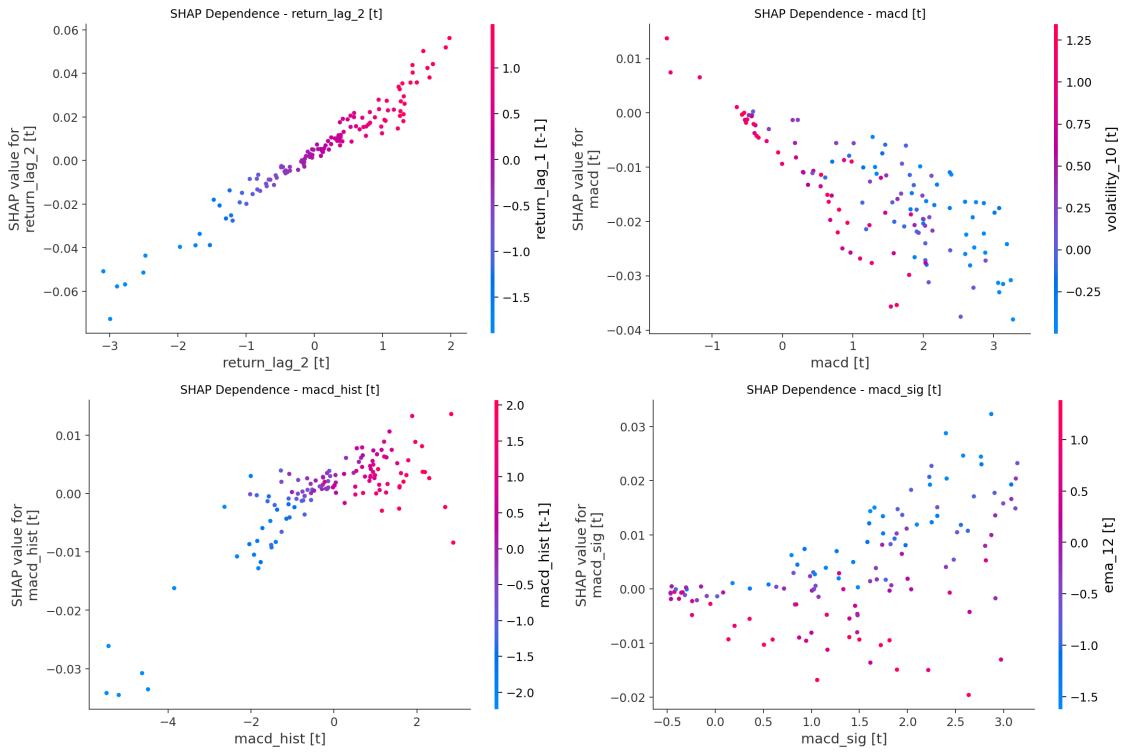


Fig. 5. Dependence plots for the four highest-ranked features. These panels show how the most recent value of each feature shifts the model output in different regions of the feature space.

V. DISCUSSION

Several conclusions follow from the experiment. First, both recurrent architectures are able to extract a weak directional signal from the engineered XAUUSD feature set. The ROC AUC values above 0.55 are small but non-trivial in a noisy financial setting. Second, the GRU slightly outperforms the LSTM in AUC while using a simpler gating mechanism, which may indicate that the available training sample is not large enough to exploit the additional flexibility of the LSTM. Third, the SHAP analysis suggests that recent returns and MACD-derived momentum features dominate the learned

decision rule, while broader volatility and volume features are comparatively less important.

However, the results also define clear limitations. The predictive margin above chance is modest, and the study does not evaluate economic value after transaction costs, slippage, execution delay, or risk constraints. The model is trained on a single feature file rather than a richer multimodal information set, and no statistical significance testing across rolling re-train windows is performed. As a result, the current system should be interpreted as an interpretable baseline study rather than a productiongrade trading model.

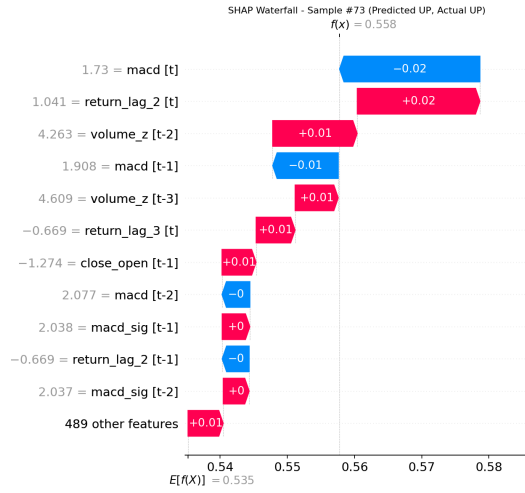


Fig. 6. Waterfall explanation for one correctly classified positive test example. The plot decomposes the final prediction into additive feature contributions and illustrates how a small number of recent signals drive the output toward the positive class.

VI. CONCLUSION

This paper presented an IEEE conference-style empirical study of LSTM and GRU baselines for daily XAUUSD directional forecasting using a prepared feature matrix of technical and return-based indicators. A chronological 70/15/15 split with 20-day rolling windows was used to produce leakage-aware train, validation, and test sets. The GRU achieved the best validation ROC AUC of 0.5488 and a test ROC AUC of 0.5570, while the LSTM achieved a slightly higher test accuracy of 0.5760. SHAP analysis showed that the most important inputs were concentrated in recent lagged returns and MACD-related variables.

The main research conclusion is therefore cautious: recurrent baselines can detect a limited directional structure in the engineered XAUUSD dataset, but the current edge is too small to justify strong practical claims. Future work should investigate stronger baselines such as temporal convolutional networks and transformers, probability calibration, walk-forward retraining, cost-aware backtesting, regime segmentation, and alternative target definitions to determine whether the observed signal is stable and economically meaningful.

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